Weak forms and problem setup — Piecewise 1D heat conduction with parameterized conductivity and source

1 Governing equations

1.1 Geometry and boundary conditions

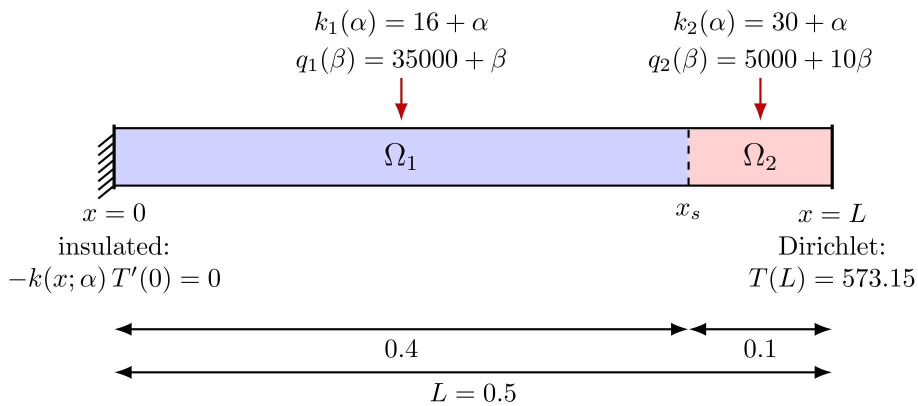

Figure 1: One-dimensional two-region heat-conduction domain with regionwise parameterized conductivity and volumetric source.

Let the computational domain be

$$\Omega = (0, L), \qquad L = 0.5.$$

The material interface is located at

$$x_s = 0.4,$$

so that

$$\Omega_1 = (0, x_s), \qquad \Omega_2 = (x_s, L).$$

The boundary conditions are

$$-k(x;\alpha)\,\frac{\mathrm{d}T}{\mathrm{d}x}(0) = 0, \qquad T(L) = T_D = 573.15.$$

The left boundary condition is natural, while the right boundary condition is an essential Dirichlet constraint.

1.2 Parameterization

The model depends on two parameters,

$$\alpha := k_{\text{param}}, \qquad \beta := q_{\text{param}},$$

with ranges

$$\alpha \in [-4, 4], \qquad \beta \in [-1000, 1000].$$

The regionwise conductivities are

$$k_1(\alpha) = 16 + \alpha, \qquad k_2(\alpha) = 30 + \alpha,$$

and the regionwise volumetric heat sources are

$$q_1(\beta) = 35000 + \beta, \qquad q_2(\beta) = 5000 + 10\beta.$$

Thus,

$$k(x;\alpha) = \begin{cases} k_1(\alpha), & x \in \Omega_1, \\ k_2(\alpha), & x \in \Omega_2, \end{cases} \qquad q(x;\beta) = \begin{cases} q_1(\beta), & x \in \Omega_1, \\ q_2(\beta), & x \in \Omega_2. \end{cases}$$

1.3 Strong form

Find the temperature field $T : \Omega \to \mathbb{R}$ such that

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left(k(x;\alpha)\frac{\mathrm{d}T}{\mathrm{d}x}\right) = q(x;\beta) \qquad \text{in } \Omega, \tag{1}$$

$$-k(0;\alpha)\frac{\mathrm{d}T}{\mathrm{d}x}(0) = 0 \qquad \text{at } x = 0, \tag{2}$$

$$T(L) = T_D = 573.15 \qquad \text{at } x = L. \tag{3}$$

No essential condition is imposed at the left endpoint because the zero-flux condition is natural.

2 Weak form

2.1 Variational statement

Define the trial and test spaces by

$$V_D = \{\, T \in H^1(\Omega) : T(L) = T_D \,\}, \qquad V_0 = \{\, v \in H^1(\Omega) : v(L) = 0 \,\}.$$

Multiplying the strong form by $v \in V_0$, integrating over Ω, and integrating by parts gives

$$\int_\Omega k(x;\alpha)\frac{\mathrm{d}T}{\mathrm{d}x}\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x - \left[k(x;\alpha)\frac{\mathrm{d}T}{\mathrm{d}x}v\right]_0^L = \int_\Omega q(x;\beta)v\,\mathrm{d}x.$$

The boundary term vanishes because $v(L) = 0$ and the left boundary has zero flux. Hence the weak problem is:
Find $T \in V_D$ such that

$$a(T, v; \alpha) = \ell(v; \beta) \qquad \forall v \in V_0,$$

where

$$a(T, v; \alpha) = \sum_{r=1}^{2} \int_{\Omega_r} k_r(\alpha)\,\frac{\mathrm{d}T}{\mathrm{d}x}\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x, \tag{4}$$

and

$$\ell(v; \beta) = \sum_{r=1}^{2} \int_{\Omega_r} q_r(\beta)\,v\,\mathrm{d}x. \tag{5}$$

2.2 Affine decomposition

The affine structure is explicit in both the bilinear and linear forms. Define the parameter-independent regional operators by

$$a_r(T, v) = \int_{\Omega_r} \frac{\mathrm{d}T}{\mathrm{d}x}\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x, \qquad \ell_r(v) = \int_{\Omega_r} v\,\mathrm{d}x, \qquad r = 1, 2.$$

Then

$$a(T, v; \alpha) = (16 + \alpha)a_1(T, v) + (30 + \alpha)a_2(T, v),$$

and

$$\ell(v; \beta) = (35000 + \beta)\ell_1(v) + (5000 + 10\beta)\ell_2(v).$$

In matrix form, let $\mathbf{K}_r$ and $\mathbf{f}_r$ denote the regional stiffness matrices and source vectors. Then

$$\mathbf{K}(\alpha) = (16 + \alpha)\mathbf{K}_1 + (30 + \alpha)\mathbf{K}_2, \tag{6}$$

and

$$\mathbf{f}(\beta) = (35000 + \beta)\mathbf{f}_1 + (5000 + 10\beta)\mathbf{f}_2. \tag{7}$$

Therefore, for every parameter pair (α, β), the full-order model is

$$\mathbf{K}(\alpha)\mathbf{T} = \mathbf{f}(\beta), \tag{8}$$

with the Dirichlet value imposed at the right boundary.

Region	Conductivity	Source	Spatial interval
Ω_1	$16 + \alpha$	$35000 + \beta$	(0, 0.4)
Ω_2	$30 + \alpha$	$5000 + 10\beta$	(0.4, 0.5)